ITAI 1371

Module 13 – Reflective Journal

Student: Alexander DeBusk

Individual submission follows

L13 – Reflection – Alexander DeBusk

In this lab, I learned how to use Scikit-learn Pipelines to organize the different steps of a machine learning workflow. I compared the manual approach, where preprocessing and model training are handled separately, with the Pipeline approach, which combines everything into one object.

One of the most important things I learned was how Pipelines help prevent data leakage. By placing preprocessing steps inside the Pipeline, the transformations are fitted using only the training data. This prevents information from the test data from affecting the training process. I also learned how to use make_pipeline and make_column_transformer to apply different preprocessing steps to numerical and categorical features.

Another useful part of this lab was understanding how additional steps, such as PCA, can be added to a Pipeline before the machine learning model. This makes it easier to expand and modify a workflow without having to manage every step separately.

This lab helped me understand why Pipelines are useful for creating cleaner, more organized, and reproducible machine learning workflows. They also make it easier to share and deploy models because the preprocessing steps and trained model can all be kept together in one object.

ITAI 1371

Module 13 – Reflective Journal

Student: Shareefah Adeola

Individual submission follows

Reflective Journal – Module 13: Building Machine Learning Pipelines

ITAI 1371

Student: Shareefah Adeola

Date: November 13, 2026

This module helped me understand the importance of Machine Learning Pipelines in developing efficient, reliable, and reproducible machine learning models.

I learned the difference between manual preprocessing and using a Pipeline approach. In the manual method, I performed individual operations such as scaling numerical features, encoding categorical variables, combining processed data, and training the classification model. In the Pipeline method, I combined these operations into a single workflow using make_pipeline and make_column_transformer.

One important concept I learned was data leakage and why it is necessary to use fit_transform() on training data and only transform() on testing data. I also learned how PCA (Principal Component Analysis) can be incorporated into a Pipeline to reduce dimensionality.

During the practical exercise, I encountered some coding errors, particularly with missing imports, which I resolved by correcting the code and rerunning the notebook.

Both the manual and Pipeline methods achieved an accuracy of 82.68% in my exercise. This helped me understand that the main advantage of a Pipeline is not necessarily higher accuracy, but better organization, consistency, and reliability.

Overall, this exercise improved my understanding of machine learning workflows and how professional data scientists organize preprocessing and model development for real-world applications.

ITAI 1371

Module 13 – Reflective Journal

Student: Yilin Leng

Individual submission follows

Reflection

In this lab, I learned how to use Scikit-learn Pipelines to organize machine learning workflows. By comparing the manual method with the Pipeline approach using the Titanic dataset, I understood how combining data preprocessing and model training makes the code cleaner and easier to maintain. One important thing I learned is that Pipelines help prevent data leakage by fitting preprocessing steps only on the training data. I also learned how to use different transformers for numerical and categorical features in one workflow. Overall, this lab helped me better understand how to build more organized and reliable machine learning models.